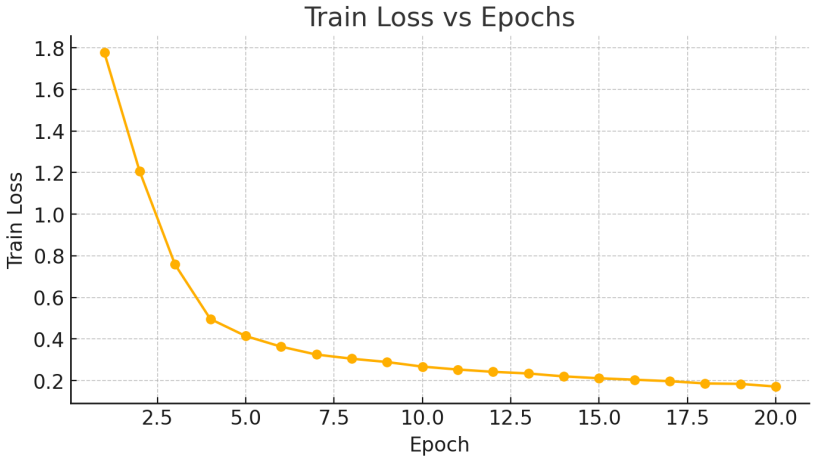
COL774 Assignment 4

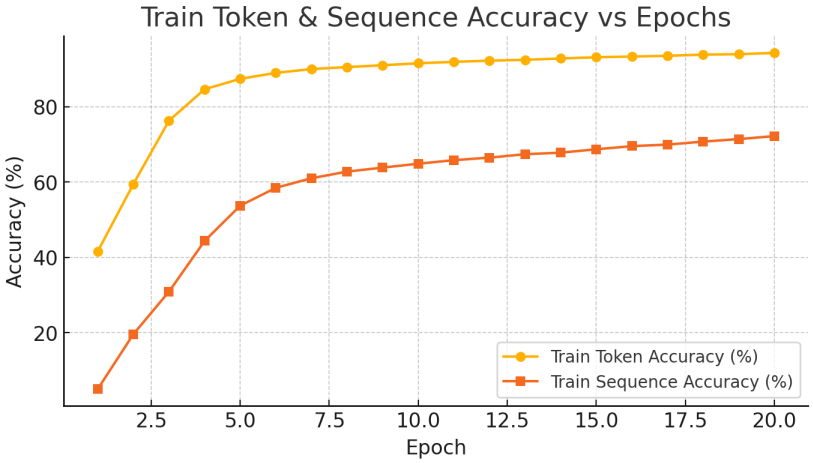
Rachit Bhalani (2023CS10961), Sabhya Arora (2023CS11177)

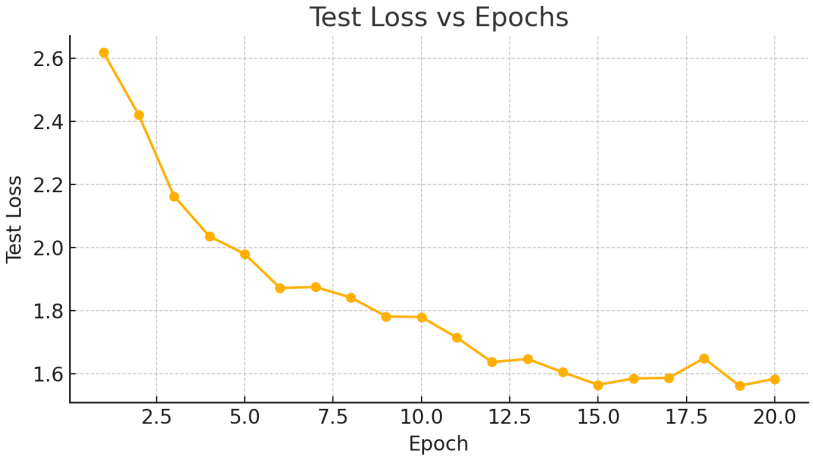
November 2025

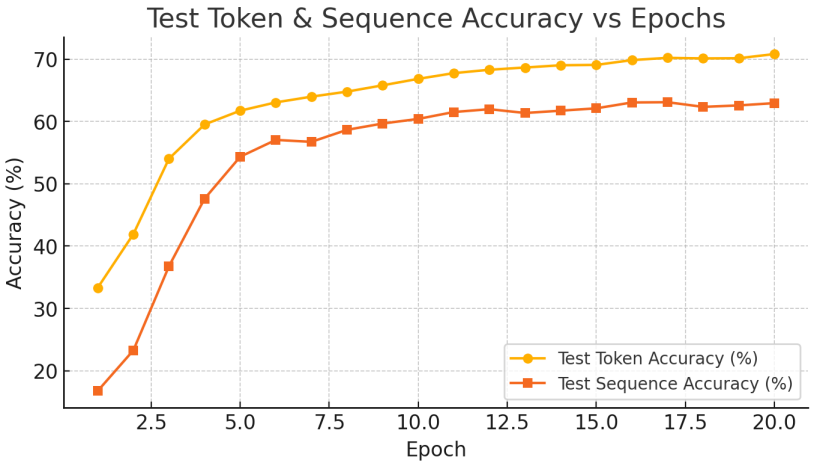
1 Recurrent Neural network

1.1 Training and test curves

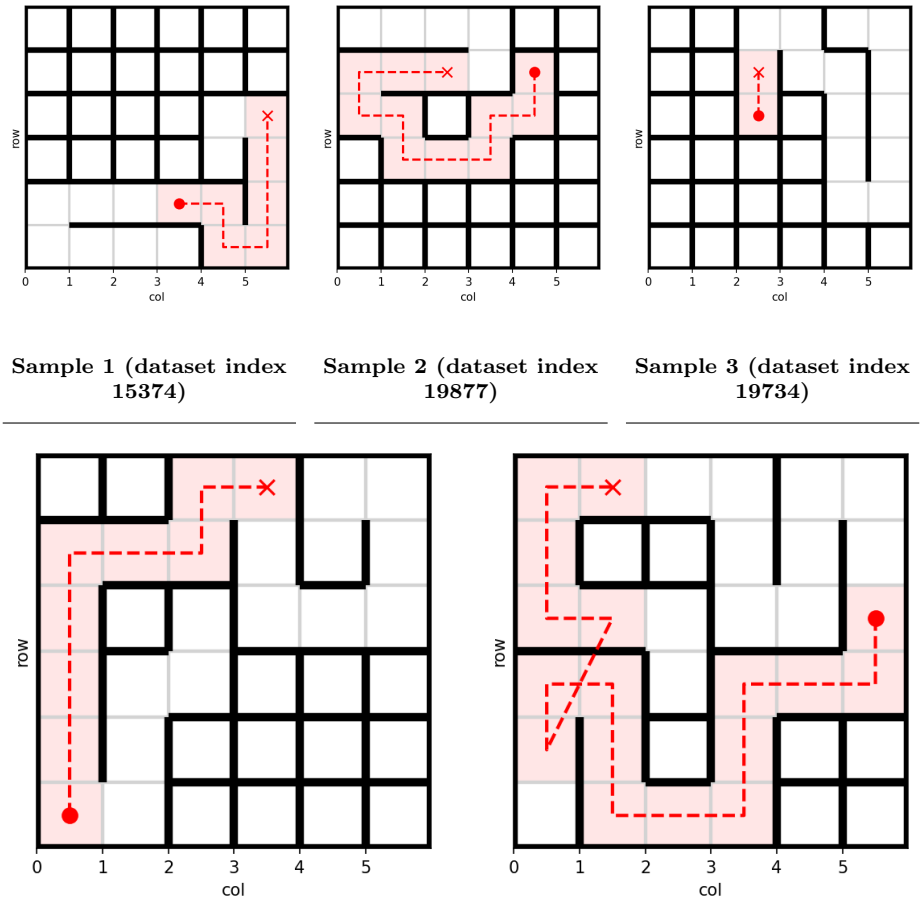



Train Token & Sequence Accuracy vs Epochs
Accuracy (%)
Epoch
Train Token Accuracy (%)
Train Sequence Accuracy (%)
20
40
60
80
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Test Loss vs Epochs
Test Loss
Epoch
1.6
1.8
2.0
2.2
2.4
2.6
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Test Token & Sequence Accuracy vs Epochs
Accuracy (%)
70
60
50
40
30
20
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Epoch
Test Token Accuracy (%)
Test Sequence Accuracy (%)

1.2 Maze predictions — 5 random validation samples



Sample 1 (dataset index 15374) | **Sample 2 (dataset index 19877)** | **Sample 3 (dataset index 19734)**

Sample 4 (dataset index 4366) | **Sample 5 (dataset index 15055)**

Figure 1: Predicted paths visualized for 5 random validation mazes.

Sample details (tokens, ground-truth, predicted, saved visualization path)

==========

Sample 1 (dataset index 15374)

```
Input tokens (trimmed PAD):
['<ADJLIST_START>', '(4,1)', '<-->', '(4,2)', ';', '(4,5)', '<-->', '(3,5)', ';',
     '(3,4)', '<-->', '(2,4)', ';', '(4,5)', '<-->', '(5,5)', ';', '(5,1)',
    '<-->', '(5,2)', ';', '(3,5)', '<-->', '(2,5)', ';', '(5,4)', '<-->', '(4,4)
    ', ';', '(5,0)', '<-->', '(5,1)', ';', '(4,0)', '<-->', '(5,0)', ';', '(4,3)
    ', '<-->', '(4,2)', ';', '(2,4)', '<-->', '(2,5)', ';', '(4,3)', '<-->',
    '(4,4)', ';', '(5,4)', '<-->', '(5,5)', ';', '(5,3)', '<-->', '(5,2)', ';',
    '(4,1)', '<-->', '(4,0)', ';', '<ADJLIST_END>', '<ORIGIN_START>', '(4,3)',
    '<ORIGIN_END>', '<TARGET_START>', '(2,5)', '<TARGET_END>', '<PATH_START>']

Ground-truth path tokens:
['(4,3)', '(4,4)', '(5,4)', '(5,5)', '(4,5)', '(3,5)', '(2,5)', '<PATH_END>']

Predicted path tokens:
['(4,3)', '(4,4)', '(5,4)', '(5,5)', '(4,5)', '(3,5)', '(2,5)', '<PATH_END>']

Visualization:
Saved maze plot to: /content/drive/MyDrive/rnn_attention/maze_plot_edac2744.png
```

==========

Sample 2 (dataset index 19877)

```
Input tokens (trimmed PAD):
['<ADJLIST_START>', '(2,1)', '<-->', '(3,1)', ';', '(3,3)', '<-->', '(2,3)', ';',
     '(1,2)', '<-->', '(1,3)', ';', '(0,0)', '<-->', '(0,1)', ';', '(2,1)',
    '<-->', '(2,0)', ';', '(0,3)', '<-->', '(0,2)', ';', '(2,4)', '<-->', '(1,4)
    ', ';', '(1,2)', '<-->', '(1,1)', ';', '(1,3)', '<-->', '(0,3)', ';', '(3,1)
    ', '<-->', '(3,2)', ';', '(0,2)', '<-->', '(0,1)', ';', '(2,3)', '<-->',
    '(2,4)', ';', '(1,1)', '<-->', '(1,0)', ';', '(3,3)', '<-->', '(3,2)', ';',
    '(2,0)', '<-->', '(1,0)', ';', '<ADJLIST_END>', '<ORIGIN_START>', '(1,4)',
    '<ORIGIN_END>', '<TARGET_START>', '(1,2)', '<TARGET_END>', '<PATH_START>']

Ground-truth path tokens:
['(1,4)', '(2,4)', '(2,3)', '(3,3)', '(3,2)', '(3,1)', '(2,1)', '(2,0)', '(1,0)',
     '(1,1)', '(1,2)', '<PATH_END>']

Predicted path tokens:
['(1,4)', '(2,4)', '(2,3)', '(3,3)', '(3,2)', '(3,1)', '(2,1)', '(2,0)', '(1,0)',
     '(1,1)', '(1,2)', '<PATH_END>', '(0,3)', '(0,2)', '(0,1)', '(0,0)', '(1,0)
    ', '(1,1)', '(1,2)', '<PATH_END>']

Visualization:
Saved maze plot to: /content/drive/MyDrive/rnn_attention/maze_plot_c85140c6.png
```

==========

Sample 3 (dataset index 19734)

```
Input tokens (trimmed PAD):
['<ADJLIST_START>', '(1,5)', '<-->', '(2,5)', ';', '(4,5)', '<-->', '(3,5)', ';',
     '(0,5)', '<-->', '(0,4)', ';', '(3,4)', '<-->', '(2,4)', ';', '(1,3)',
    '<-->', '(0,3)', ';', '(1,2)', '<-->', '(0,2)', ';', '(4,5)', '<-->', '(4,4)
    ', ';', '(1,2)', '<-->', '(2,2)', ';', '(0,5)', '<-->', '(1,5)', ';', '(2,4)
    ', '<-->', '(1,4)', ';', '(1,3)', '<-->', '(1,4)', ';', '(3,4)', '<-->',
    '(4,4)', ';', '(0,3)', '<-->', '(0,2)', ';', '(2,5)', '<-->', '(3,5)', ';',
    '<ADJLIST_END>', '<ORIGIN_START>', '(2,2)', '<ORIGIN_END>', '<TARGET_START
    >', '(1,2)', '<TARGET_END>', '<PATH_START>']

Ground-truth path tokens:
['(2,2)', '(1,2)', '<PATH_END>']

Predicted path tokens:
['(2,2)', '(1,2)', '<PATH_END>']

Visualization:
Saved maze plot to: /content/drive/MyDrive/rnn_attention/maze_plot_ba404c63.png
```

==========

Sample 4 (dataset index 4366)

```
Input tokens (trimmed PAD):
['<ADJLIST_START>', '(1,5)', '<-->', '(2,5)', ';', '(5,0)', '<-->', '(4,0)', ';',
     '(1,1)', '<-->', '(1,2)', ';', '(5,0)', '<-->', '(5,1)', ';', '(2,3)',
    '<-->', '(1,3)', ';', '(1,5)', '<-->', '(0,5)', ';', '(2,3)', '<-->', '(2,4)
    ', ';', '(0,3)', '<-->', '(1,3)', ';', '(1,4)', '<-->', '(0,4)', ';', '(1,0)
    ', '<-->', '(1,1)', ';', '(2,2)', '<-->', '(3,2)', ';', '(2,0)', '<-->',
    '(3,0)', ';', '(1,0)', '<-->', '(2,0)', ';', '(2,4)', '<-->', '(2,5)', ';',
    '(5,1)', '<-->', '(4,1)', ';', '(0,5)', '<-->', '(0,4)', ';', '(3,0)',
    '<-->', '(4,0)', ';', '(4,1)', '<-->', '(3,1)', ';', '(0,2)', '<-->', '(0,3)
    ', ';', '(0,2)', '<-->', '(1,2)', ';', '(3,2)', '<-->', '(3,1)', ';', '<
    ADJLIST_END>', '<ORIGIN_START>', '(5,0)', '<ORIGIN_END>', '<TARGET_START>',
    '(0,3)', '<TARGET_END>', '<PATH_START>']

Ground-truth path tokens:
['(5,0)', '(4,0)', '(3,0)', '(2,0)', '(1,0)', '(1,1)', '(1,2)', '(0,2)', '(0,3)',
     '<PATH_END>']

Predicted path tokens:
['(5,0)', '(4,0)', '(3,0)', '(2,0)', '(1,0)', '(1,1)', '(1,2)', '(0,2)', '(0,3)',
     '<PATH_END>']

Visualization:
Saved maze plot to: /content/drive/MyDrive/rnn_attention/maze_plot_f485af4f.png
```

==========

Sample 5 (dataset index 15055)

```
Input tokens (trimmed PAD):
['<ADJLIST_START>', '(5,0)', '<-->', '(4,0)', ';', '(1,0)', '<-->', '(2,0)', ';',
     '(0,1)', '<-->', '(0,0)', ';', '(1,3)', '<-->', '(0,3)', ';', '(4,1)',
    '<-->', '(5,1)', ';', '(2,3)', '<-->', '(1,3)', ';', '(1,0)', '<-->', '(0,0)
    ', ';', '(1,4)', '<-->', '(2,4)', ';', '(2,5)', '<-->', '(3,5)', ';', '(2,2)
    ', '<-->', '(3,2)', ';', '(1,5)', '<-->', '(2,5)', ';', '(5,1)', '<-->',
```

```
    '(5,2)', ';', '(0,4)', '<-->', '(1,4)', ';', '(2,2)', '<-->', '(2,1)', ';',
    '(4,1)', '<-->', '(3,1)', ';', '(2,1)', '<-->', '(2,0)', ';', '(3,0)',
    '<-->', '(4,0)', ';', '(0,5)', '<-->', '(0,4)', ';', '(3,0)', '<-->', '(3,1)
    ', ';', '(0,3)', '<-->', '(0,2)', ';', '(1,5)', '<-->', '(0,5)', ';', '(2,3)
    ', '<-->', '(2,4)', ';', '(3,4)', '<-->', '(3,3)', ';', '(4,3)', '<-->',
    '(5,3)', ';', '(5,2)', '<-->', '(5,3)', ';', '(3,4)', '<-->', '(3,5)', ';',
    '(4,3)', '<-->', '(3,3)', ';', '(0,1)', '<-->', '(0,2)', ';', '<ADJLIST_END
    >', '<ORIGIN_START>', '(2,5)', '<ORIGIN_END>', '<TARGET_START>', '(0,1)', '<
    TARGET_END>', '<PATH_START>']

Ground-truth path tokens:
['(2,5)', '(1,5)', '(0,5)', '(0,4)', '(1,4)', '(2,4)', '(2,3)', '(1,3)', '(0,3)',
     '(0,2)', '(0,1)', '<PATH_END>']

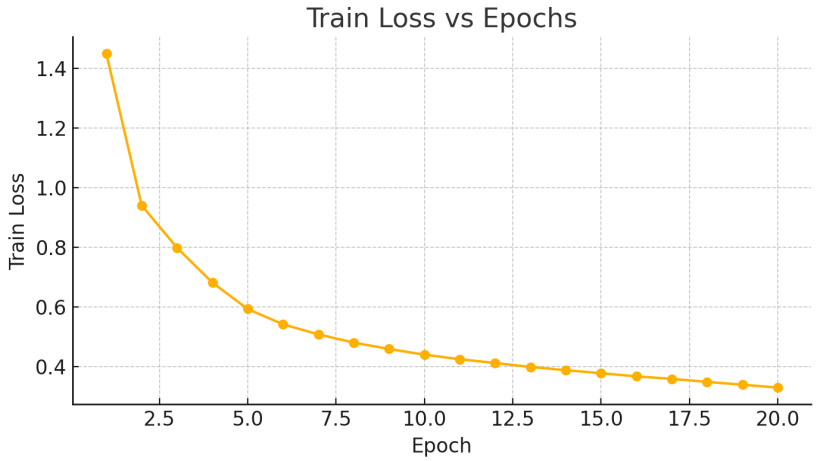
Predicted path tokens:
['(2,5)', '(3,5)', '(3,4)', '(3,3)', '(4,3)', '(5,3)', '(5,2)', '(5,1)', '(4,1)',
     '(3,1)', '(3,0)', '(4,0)', '(2,1)', '(2,0)', '(1,0)', '(0,0)', '(0,1)', '<
    PATH_END>']

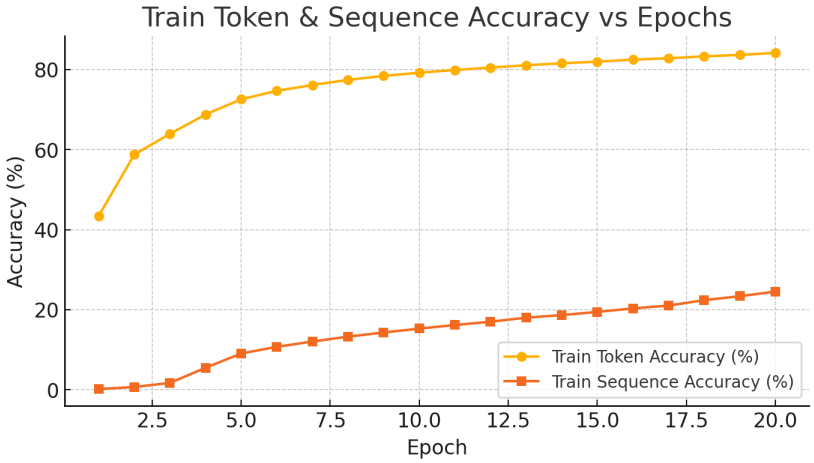
Visualization:
Saved maze plot to: /content/drive/MyDrive/rnn_attention/maze_plot_a574d79f.png
```

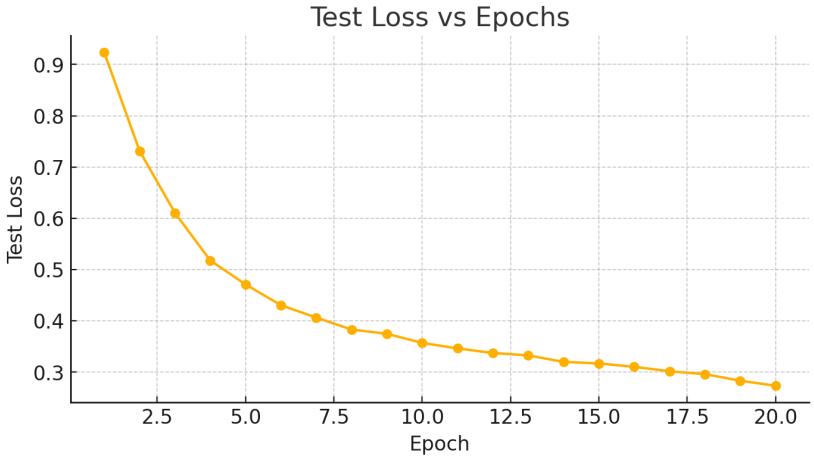
==========

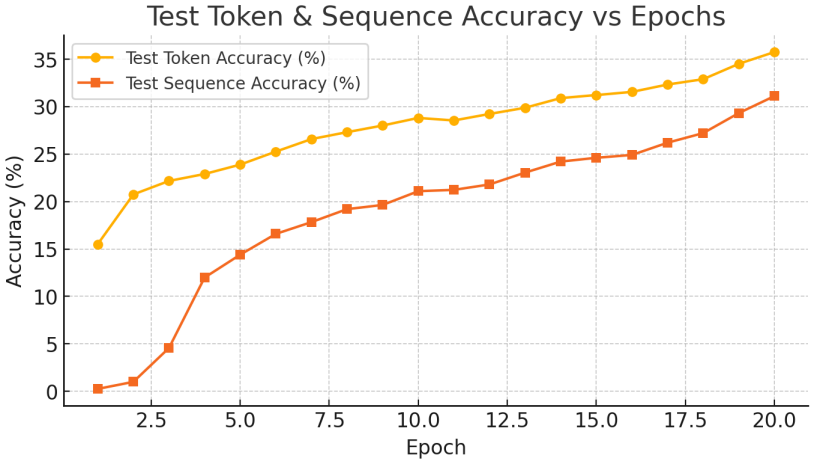
2 Transformer Architecture

2.1 Training and test curves




Train Token & Sequence Accuracy vs Epochs
Accuracy (%)
Epoch
Train Token Accuracy (%)
Train Sequence Accuracy (%)
0
20
40
60
80
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Test Loss vs Epochs
Test Loss
Epoch
0.3
0.4
0.5
0.6
0.7
0.8
0.9
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Test Token & Sequence Accuracy vs Epochs
Test Token Accuracy (%)
Test Sequence Accuracy (%)
Accuracy (%)
Epoch
0
5
10
15
20
25
30
35
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

2.2 Qualitative Results: 5 Random Validation Mazes

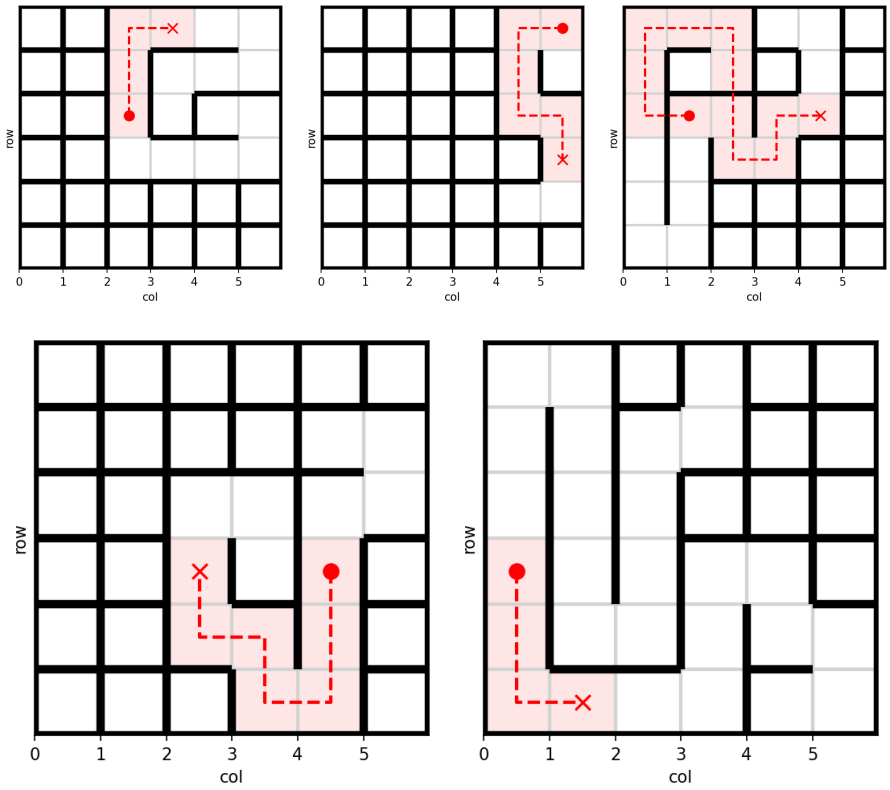


Figure 2: Visualizations of predicted paths for 5 random mazes.

Sample Details

Sample 1 (dataset index 11238)

```
Input tokens:
['<ADJLIST_START>', '(2,5)', '<-->', '(3,5)', ... '<PATH_START>']

Ground truth:
['<SOS>', '(2,2)', '(1,2)', '(0,2)', '(0,3)', '<PATH_END>']

Predicted (truncated):
['(2,2)', '(1,2)', '(0,2)', '(0,3)', '<PATH_END>']

Visualization saved as: maze_plot_25a1953b.png
```

Sample 2 (dataset index 7613)

```
Input tokens:
```

```
['<ADJLIST_START>', '(1,4)', '<-->', '(0,4)', ... '<PATH_START>']

Ground truth:
['<SOS>', '(3,5)', '(2,5)', '(2,4)', '(1,4)', '(0,4)', '(0,5)', '<PATH_END>']

Predicted (truncated):
['(0,5)', '(0,4)', '(1,4)', '(2,4)', '(2,5)', '(3,5)', '<PATH_END>']

Visualization saved as: maze_plot_3bdc174d.png
```

Sample 3 (dataset index 13977)

```
Input tokens:
['<ADJLIST_START>', '(2,0)', '<-->', '(3,0)', ... '<PATH_START>']

Ground truth:
['<SOS>', '(2,1)', '(2,2)', '(3,2)', '(3,3)', '(2,3)', '(2,4)', '<PATH_END>']

Predicted (truncated):
['(2,1)', '(2,0)', '(1,0)', '(0,0)', '(0,1)', '(0,2)',
 '(1,2)', '(2,2)', '(3,2)', '(3,3)', '(2,3)', '(2,4)', '<PATH_END>']

Visualization saved as: maze_plot_75b6dbb0.png
```

Sample 4 (dataset index 5875)

```
Input tokens:
['<ADJLIST_START>', '(5,3)', '<-->', '(4,3)', ... '<PATH_START>']

Ground truth:
['<SOS>', '(3,4)', '(4,4)', '(5,4)', '(5,3)', '(4,3)', '(4,2)', '(3,2)', '<
    PATH_END>']

Predicted (truncated):
['(3,4)', '(4,4)', '(5,4)', '(5,3)', '(4,3)', '(4,2)', '(3,2)', '<PATH_END>']

Visualization saved as: maze_plot_7fc8dd9a.png
```

Sample 5 (dataset index 5754)

```
Input tokens:
['<ADJLIST_START>', '(3,2)', '<-->', '(2,2)', ... '<PATH_START>']

Ground truth:
['<SOS>', '(3,0)', '(4,0)', '(5,0)', '(5,1)', '<PATH_END>']

Predicted (truncated):
['(3,0)', '(4,0)', '(5,0)', '(5,1)', '<PATH_END>']

Visualization saved as: maze_plot_6d54138b.png
```

3 Comparison

Although both models appear similar in the qualitative visualizations—where 4 out of 5 randomly chosen samples are correct—this observation is not representative of their true performance. The samples for the Transformer happened to

include several correctly solved mazes purely by chance, making its qualitative results appear stronger than they actually are. Quantitative evaluation over the full validation set provides a more reliable comparison, and these metrics show that the RNN achieves consistently higher token-level and sequence-level accuracy than the Transformer. Therefore, despite similar-looking visual examples, the RNN demonstrates clearly superior overall performance.